Cofactor-aware algebraic oracle (pkm-semaev lifts=1)

Cofactor-aware algebraic oracle (`pkm-semaev:lifts=1`) on the k = 20 Curve25519 miniature

Protocol: PROTOCOL.md (frozen before data; cells, seeds and rules unchanged). Everything below is scoped to k = 20 and the five cells measured; nothing is generalised. F4 multiply-adds are native, unpriced work (AGENTS.md section 5a). No rho arm, no cross-method claim, no leaderboard row.

What was built

- `src/cryptanalysis/ic_framework/pkm_semaev.rs`: parameter `lifts=1`. `prepare` computes the rational h-torsion `E(F_q)[h]` with `cofactor_fiber` (charged to set-up); `decompose` tries `R + K` for each K whose lift abscissa has not yet been tried (`K = O` first), one F4 system per lift, first liftable relation wins; every lift addition and every F4 call is charged. `setup_native()` reports fiber size, cofactor, completeness and samples. With h = 1 it is the blind oracle.
- `src/bin/curve25519_family.rs`: `f4cost ... <cap_s> 2 full <0|1>` reports `lifts`, `hits_exact`, `setup_group_ops`, `setup_native`.
- `run_cells.sh`: shell runner (`ulimit -v 3500000`, `timeout 900`, per-cell stderr and `.status`), appends to `analysis/f4cost_lifts.jsonl` (7.4 KB, committed).

The lift code compiled as written (lib and bin) with no source fix. The one defect found was in the new unit test, not in the oracle (see Deviations).

Unit test

`cargo test --release --lib pkm_semaev`: `test result: ok. 10 passed; 0 failed; 0 ignored; 0 measured; 6054 filtered out; finished in 542.29s` (the 9 existing tests plus `cofactor_lifts_match_the_brute_force_fiber`). Counts printed by the new test (h = 4 curve over the 10007 field, 80 random subgroup targets): blind-decomposable 3, lift-decomposable 21 (brute force over the torsion found by enumeration); 36 relations from a non-identity lift summed over the two lifted oracles (guess = 0 and guess = 1). All verdicts equalled the brute-force sets; every relation had `Sum - R` in E[h], `[h]Sum = [h]R` and satisfied the pipeline row `sum coef*x = h*a (mod r)`; h = 1 gave identical relations with and without `lifts=1`; extra F4 calls were charged. The test prints that line with `eprintln!` (added to the WIP test).

Measurement

Binary `/home/user/frozen-bin/curve25519_family_lead2`, sha256 `9aa09afccbc9f59bc302252db81dc1e6a99becfc5e98a810b438f80a266a97bc`, built from commit ad50d0892f plus the uncommitted test fix. One process at a time on one host, pairs interleaved (lifts=0 then lifts=1, same seed). All 10 runs finished (`.status` all ok), none capped (longest 281 s of the 600 s cap), `undecided` = 0 everywhere. Fiber: h = 8, size 8, complete, in every lifted cell.

form	t	guess	arm	targets	hits	hits_exact	hits verified	F4 calls/target	add
isogeny	4	0	lifts=0	200	0	0	0	1.0	
isogeny	4	0	lifts=1	200	1	0	1	7.995	
isogeny	5	0	lifts=0	200	0	0	0	1.0	
isogeny	5	0	lifts=1	200	0	0	0	8.0	
isogeny	6	0	lifts=0	200	0	0	0	1.0	
isogeny	6	0	lifts=1	200	0	0	0	8.0	
bits	5	3	lifts=0	200	0	0	0	8.0	
bits	5	3	lifts=1	200	0	0	0	64.0	
bits	6	3	lifts=0	200	0	0	0	8.0	
bits	6	3	lifts=1	200	4	0	4	63.18	

Ratios lifts / no-lifts (exact counts): F4 calls per target x7.995, x8.000, x8.000, x8.000, x7.898 (cells in table order); mul-adds per target x7.995, x8.000, x8.000, x8.000, x7.898. Mul-adds per relation ratio: undefined in every cell (the no-lifts arm has 0 hits in all five).

Hit rates, Poisson (exact) 95% intervals on the count, per 200 targets: 0 hits [0, 3.69] (rate 0 to 1.8%); 1 hit [0.03, 5.57] (0.01% to 2.8%); 4 hits [1.09, 10.24] (0.5% to 5.1%). Pooled over the five cells: no-lifts 0 / 1000 targets, lifts 5 / 1000. Descriptive and post hoc (not preregistered): if the 5 pooled hits were split evenly between arms, all 5 falling in the lifts arm has two-sided probability 2 x (1/2)^5 = 0.0625. Exact projected coverage lambda of the factor bases (support from `covered_points / r`, in the jsonl): 1.4e-4, 5.5e-4, 1.3e-3 (isogeny t = 4, 5, 6), 3.2e-4, 1.2e-3 (bits t = 5, 6), i.e. 0.03 to 0.25 expected exact hits per 200 targets and at most 8x that with lifts; the cells are underpowered, as the protocol stated. All 5 lifted hits came from a non-identity K (`pkm_lift_hits_nonidentity` 1 and 4), so `hits_exact` = 0; each passes the pipeline rule `[8]Sum = [8]R`.

Figure: analysis/lifts.svg (mul-adds per target and per hit, both arms; n/a where an arm has no hit).

Predictions

- **L1** (hit rate up 3-5x, never above 8x): not testable. The no-lifts arm has 0 hits in every cell, so no factor can be formed; the lifts arm gave 5 hits in 1000 targets versus 0 in 1000 (descriptive p = 0.06, above). Neither confirmed nor refuted.
- **L2** (F4 calls per target rise at most 4x): refuted in these cells. The rise is x7.9 to x8.0 in all five (nearly every target is refuted, so all 8 lifts are tried).

- **L3** (per-relation cost ratio lifts / no-lifts): undefined in every cell; no statement possible. The per-target cost ratio is x7.9 to x8.0, so a per-relation gain would require the hit-rate gain to exceed about 8x, which is the cap L1 itself states.

Classification (preregistered rule)

Not an improvement: the criterion needs at least 3 hits in both arms in a cell, and no cell has even 1 hit in the no-lifts arm, so the per-relation ratio is undefined everywhere and cannot be below 1. Of the two remaining labels, the data show the cost side exactly (about 8x F4 work per target) and a hit-rate rise only at the level of 5 versus 0 hits, so I label it “yield gain at a cost, unproven”: hit count 0 to 5, per-target cost x8. No per-relation gain is measured. Gap to matched rho: not measured here, unchanged from the ladder study.

Deviations

1. Compile: the WIP was reported as never compiled, but `cargo check --release --lib` and `--bin curve25519_family` passed with no source change.
2. Unit test fix: the first run of `cofactor_lifts_match_the_brute_force_fiber` failed at the row assertion (`left 677, right 1626`). Cause: the test took one discrete log per factor-base point, but the base lists P (coef 1) and -P (coef r-1) on one column, so the negative point’s coefficient was applied to an already negated log. Fixed in the test only: one log per column, taken from the coef-1 member, indexed by `col_of`. The lift code itself needed no change. Also added the `eprintln!` count line.
3. The measured binary was built from the working tree (HEAD ad50d0892f + the test edits, which do not affect the binary’s oracle or CLI code).
4. `scripts/checkpoint.sh init` worked; `.resume/RESUME.md` was filled and `checkpoint.sh push` uploaded to S3 before launch. `.resume/` is not part of the research commit.
5. Runs were local (not isolab-eks), as instructed by the task; counts are operation counts, independent of load.
6. No supplementary larger-target run was done.
7. Clippy: `cargo clippy --release --lib --bin curve25519_family -- -D warnings` exit 0 (only the environmental unknown-lint `clippy::chunks_exact_to_as_chunks` warning).

Report PDF: REPORT.pdf.